Order-neighbor divisors do not mark the eleven skipped curves

Date: 2026-10-23 UTC. **Scope:** The eleven omitted prime-field seed classes and eight published same-field controls, each identified by its exact EC1 UID in analysis.json. **Verdict within this screen:** No shared auxiliary-input divisor opportunity separates the labels. This result does not test ordinary ECDLP hardness or reconstruct a 1997 selection rule.

Why test these divisors?

Cheon's auxiliary-input algorithm (`retrieved`, primary; downloaded PDF SHA-256 `fd7746b9d2c37c228b63e9b993e7eb58a90045740f5d1fd6aac3e5a5af70b4de`) is sensitive to divisors of the prime subgroup order `n−1` or `n+1`. The `n−1` route requires a point encoding the *unknown secret raised to a chosen power*, in addition to the usual public point. The `n+1` route requires a sequence of such auxiliary powers. The standard public inputs to an ECDLP, ECDSA signature, or ECDH exchange do not include these objects. The algorithm appeared in 2006, after the curve-selection period; there is no evidence that its criteria were screened in 1997.

The full `n−1` and `n+1` factorizations had already been completed and verified for these nineteen curves in the order-certificate study (`internal`). This study asks a narrower arithmetic question: do the omitted curves consistently have divisors nearer the theoretical bounds than published same-field controls? The protocol froze the bounds, inputs, cap, controls, and interpretation before this calculation.

Exact result

analyze.py read all 38 previously verified factorization outputs, checked each product against the archived `n±1`, enumerated every divisor (maximum observed count **10,240**, below the two-million cap), and selected the largest divisor at the exact integer bound. It checked the integer-root implementation with `n=101` against brute-force divisor lists and rejected injected wrong factors and a wrong group order. The exact source digests, group orders, divisors, rational gap ratios, EC1 UIDs, and source-output digests are in analysis.json (`internal`, derived). The script did not rerun factorization or point counting; primality and curve-order evidence are inherited from the cited prior studies.

Frozen opportunity metric	Omitted curve more favorable	Published control more favorable	Ties
Largest `d | n−1` at `d ≤ ⌊√n⌋`, smaller bound/divisor gap	5/25	**20/25**	0
Largest `d | n+1` at `d ≤ ⌊∛n⌋`, smaller gap	12/25	13/25	0
Largest `d | n+1` at `d ≤ 1024`, larger `d`	11/25	14/25	0

The 25 pairs compare candidates only within the four fields that have both labels. No metric has a universal direction. In P-256, the published curve's `n−1` divisor lies very close to √n (display gap **0.031 bits**), whereas omitted block 876 has a **113.722-bit** display gap. The omitted P-224 block 1485 reverses the broad trend with a **0.081-bit** gap, while the published P-224 gap is **83.009 bits**. These are approximate chart coordinates derived from exact rational ratios, not attack-cost measurements. The pair scores are exploratory: the labels and earlier factorization results were known before this study, and the stream endpoints are selected rather than random samples.

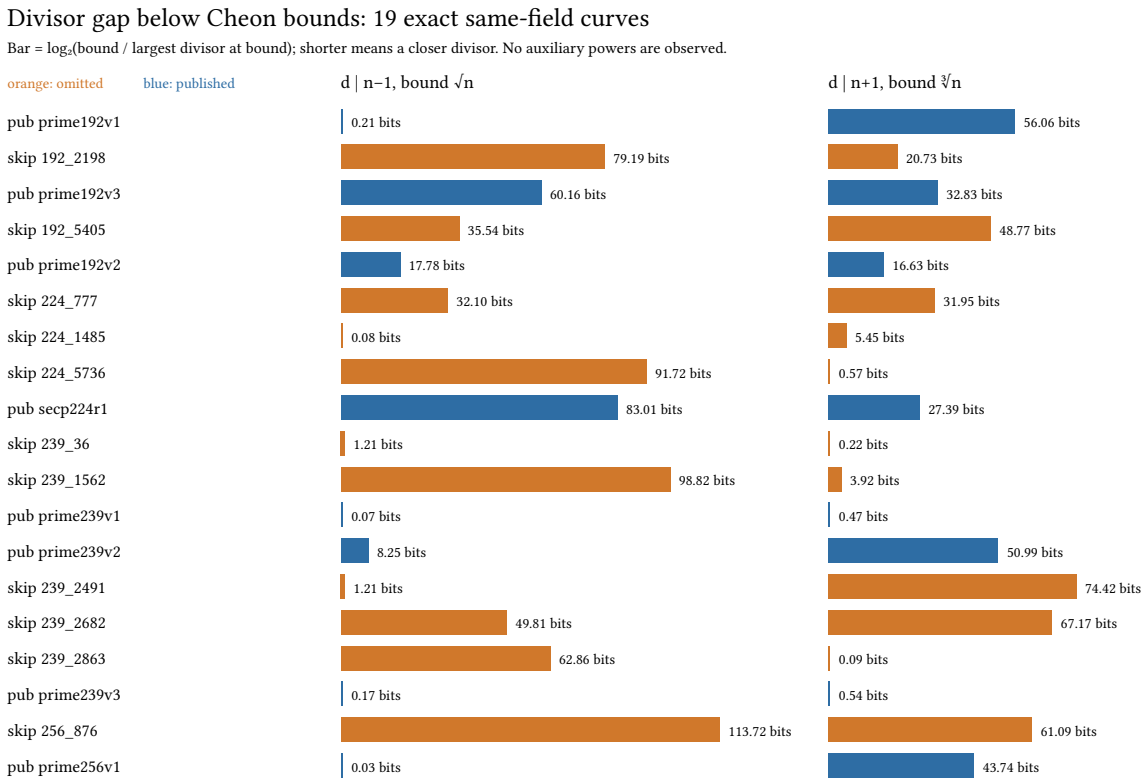


Figure 1: Exact divisor-gap comparison

Figure 1. Display `log₂(bound/largest divisor)` by curve and direction; source integers and EC1 UIDs are in analysis.json. The 1024-limited metric is tabulated in the same JSON. Editable chart source: render_visuals.py. The bars are deterministic arithmetic, so no statistical uncertainty interval applies.

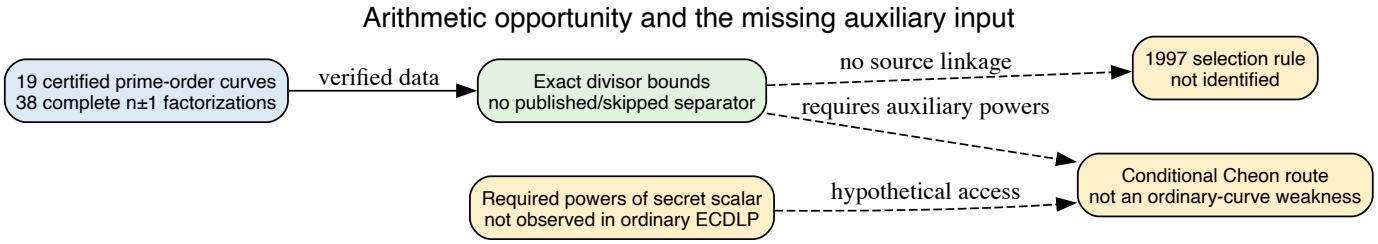


Figure 2: Inference boundary

Figure 2. The solid edge is exact arithmetic from verified factorizations. Dashed edges are missing auxiliary inputs and absent historical linkage. Editable source: inference.dot.

Interpretation and next discriminator

Derived: A rule that classifies all eleven omissions as weak solely because their orders offer more favorable divisors for this auxiliary-input route fails on published same-field controls. This closes the three declared divisor-opportunity metrics on these nineteen curves. It does not exclude other subgroup-order properties, attacks with different auxiliary data, or a hidden criterion unrelated to `n±1`. The apparent published-side `n−1` tendency is not a protective 1997 screen; it also has five reversed pairs and no contemporaneous source.

Missing premise for an attack: No protocol transcript or implementation here supplies `[x^d]G` or the longer power sequence for an unknown `x`. Even a perfect divisor match would therefore be insufficient to call an ordinary ECDLP instance weak. A proposed auxiliary-input attack must name the exact protocol that exposes the powers, the attainable `d`, the data and memory requirement, and the full cost against a same-field published control. A new quantitative speedup claim would require the repository's progress-fidelity record and independent verification.

Historical next step: A dated 1997–98 curve-selection artifact or original generator trace remains necessary to attribute any rule. The selection problem stays open. The seed-selection and canonical ECDLP graphs were checked; this screen adds no verified ordinary-DLP attack edge, so they remain unchanged. The PDF includes both figures.

Reproduce the derived screen

From the repository root run `python3 research/nist_seed_aux_input_divisors_20261023/analyze.py`, then `python3 research/nist_seed_aux_input_divisors_20261023/render_visuals.py`. Build the PDF from this directory with `pandoc report.md --metadata-file=pdf_metadata.yaml --pdf-engine=typst -o report.pdf`. The original 38 factorization verification receipt and exact curve archive (`internal`) are prerequisites. No official research state changes follow from this retrospective structural screen.